Credit Risk Analytics

An End-to-End Machine Learning Pipeline
for Consumer Lending Default Prediction

This report documents the development, evaluation, and business simulation of a machine learning model for predicting loan default at the lending platform The work was conducted using the Home Credit Default Risk dataset and represents an end-to-end credit risk analytics pipeline.

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1. Executive Summary

the lending platform is a mid-sized digital consumer-lending platform that originates unsecured personal loans of $1,000 to $35,000 with 12 to 60-month terms. The company targets the near-prime segment (credit scores 580-680), a population underserved by traditional banks but carrying an elevated default risk of approximately 8%. This report presents the development of a machine learning-based Probability of Default (PD) model designed to improve underwriting decisions, reduce credit losses, and optimize portfolio profitability.

Using the Home Credit Default Risk dataset (307,511 applicants), we engineered 182 predictive features and trained five candidate models. Our selected model — LightGBM — achieved an AUC of 0.768 on a held-out test set, with a Gini coefficient of 0.536 and recall of 69.7% at the optimal decision threshold of 0.485.

The business impact is substantial. In a simulated portfolio of 50,000 loans ($625 million total principal), the AI model at its optimal threshold generates $112.6 million in net profit with a 26.83% risk-adjusted return, approving 67% of applicants. This represents a $104.5 million (1,298%) improvement over the current rule-based underwriting policy, which approves only 4.6% of applicants and generates just $8.1 million in profit. The model's decision threshold can be adjusted across a wide range to accommodate different business cycles without retraining.

Key Findings

- Best model: LightGBM (AUC 0.768), selected from five candidates
- Dominant risk driver: EXT_MEAN — composite external creditworthiness (SHAP importance 0.47)
- AI model generates $104.5M more profit than current rules — a 1,298% improvement
- Flexible threshold permits tuning for growth or capital preservation across business cycles
- Top-5% collections intervention yields $52,575 net benefit after costs

2. Business Context

2.1 the lending platform — Company Profile

the lending platform is a digital consumer-lending platform operating in the United States. The company offers unsecured personal loans from $1,000 to $35,000 with terms of 12 to 60 months. Borrowers apply online and receive decisions within minutes. the lending platform's active portfolio comprises approximately 50,000 loans with total exposure exceeding $600 million.

the lending platform targets the near-prime segment — borrowers with credit scores between 580 and 680. This population is underserved by traditional banks, which typically require FICO scores above 680 for unsecured lending. However, near-prime borrowers carry higher default risk, making accurate risk assessment essential for sustainable profitability.

2.2 The Near-Prime Lending Opportunity

The near-prime consumer lending market in the United States is estimated at over $200 billion annually. These borrowers are often creditworthy but have limited credit histories, past credit events, or lower incomes that disqualify them from prime lending. A lender that can accurately differentiate risk within this segment can capture a large, profitable market while managing losses.

2.3 Why Default Prediction Matters

Accurate default prediction delivers value across the lending lifecycle:

- Origination: Flag high-risk applicants before funding, enabling declines or risk-adjusted terms
- Capital reserves: Better prediction reduces regulatory capital requirements under Basel III
- Pricing: Risk-segmented pricing retains good customers while protecting margins
- Collections: PD scores prioritize collection resources on accounts most likely to default

A 5-point improvement in AUC (e.g., 0.75 to 0.80) is estimated to reduce net charge-offs by 15-20%, worth approximately $2-3 million annually for a portfolio of this size.

2.4 Stakeholder Map

Stakeholder	Role	Key Question
Risk Management	Credit risk oversight	Decision quality?
Underwriting Team	Makes loan decisions	Can we trust this score to automate decisions?
Collections Manager	Manages delinquent accounts	Which accounts should we call first?
Finance / Treasury	Manages capital allocation	What is our expected loss next quarter?
Regulatory Compliance	Ensures fair lending	Does the model discriminate?
Product Manager	Designs loan terms	Can we offer better rates to good borrowers?

Table 1: Stakeholder map for the credit risk analytics initiative

2.5 Success Criteria

The project's success was evaluated against both technical and business criteria:

- Primary — Technical: Build a default-probability model achieving AUC >= 0.80 on a holdout test set
- Secondary — Business: Translate model outputs into dollar-denominated profit simulations
- Tertiary — Deployment: Produce dashboard exports for stakeholder consumption
- Compliance: Perform fairness analysis to identify any disparate impact

3. Data Overview & Exploratory Analysis

3.1 Dataset: Home Credit Default Risk

The Home Credit Default Risk dataset from Kaggle contains 307,511 loan applicants with 122 raw features from a genuine consumer-lending company. The dataset was selected for its realistic imbalance (~8% default rate), richness of applicant-level data, and industry relevance.

Table 2: Dataset summary

Attribute	Detail
Source	Kaggle: Home Credit Default Risk
Applicants	307,511 (80% train / 20% test)
Raw Features	122 columns
Target	TARGET — 1 if defaulted (90+ DPD), 0 if repaid
Default Rate	8.07%

3.2 Target Variable & Base Rates

The target variable TARGET indicates whether a borrower defaulted on their loan, defined by Home Credit as having a payment delay of more than X days on the first installment — consistent with the industry-standard 90+ days past due definition. The base default rate is 8.07%, yielding an imbalance ratio of 11.4:1 (majority to minority class). This means a naive "predict all 0" model would achieve 92% accuracy while being completely useless for risk differentiation.

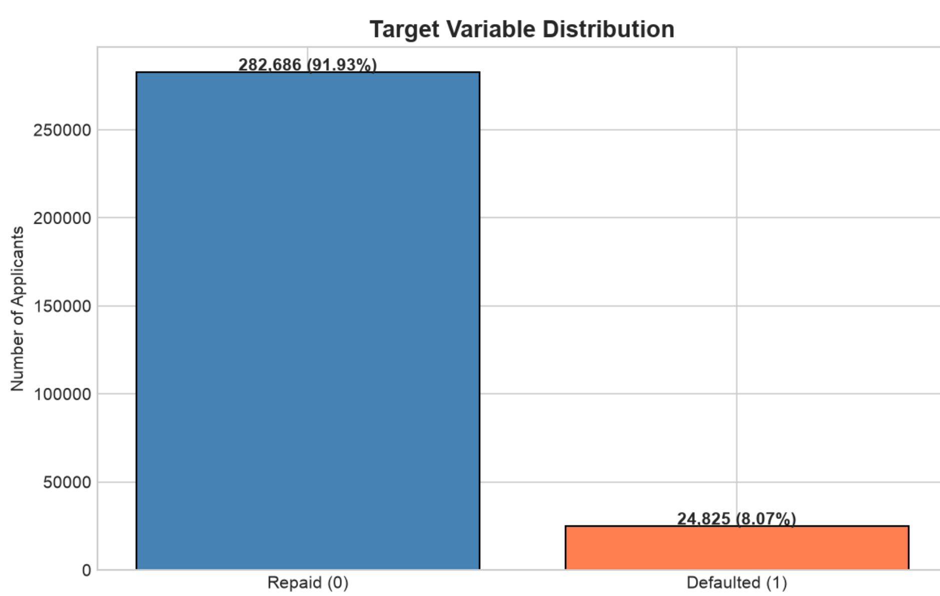

Figure 1: Target distribution showing 8.07% default rate

3.3 Data Quality Assessment

A comprehensive data quality check revealed several challenges requiring remediation:

- Missing values: Many features had missingness exceeding 60%, particularly external credit sources
- DAYS_EMPLOYED artifact: 55,374 applicants (18.0%) showed a value of 365,243 days, encoding unemployed status rather than actual employment duration
- Outliers: Extreme values in income (AMT_INCOME_TOTAL) and loan amount (AMT_CREDIT) required domain-informed capping
- No duplicate records found — the primary key SK_ID_CURR is unique
- The DAYS_EMPLOYED artifact is informative: unemployed applicants show 5.40% default rate vs. 8.66% for employed applicants

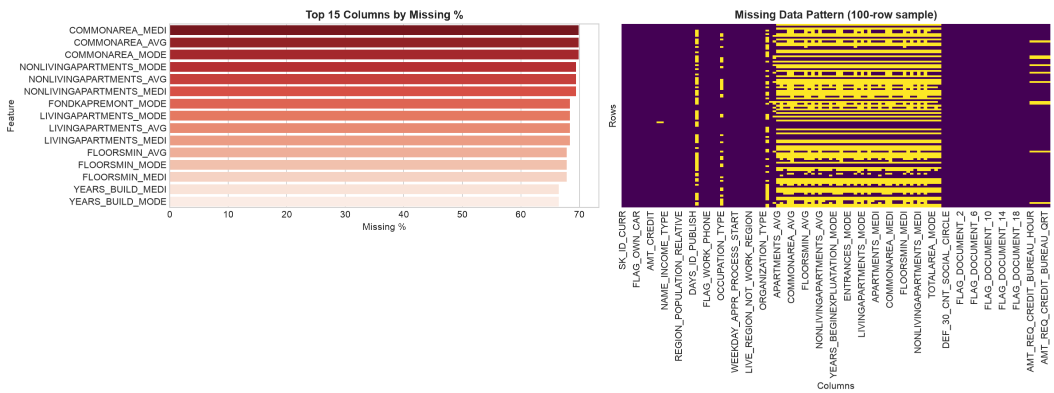

Figure 2: Missing value analysis across features

3.4 Risk Driver Identification

Exploratory data analysis identified several powerful risk signals:

External Credit Scores

- EXT_SOURCE_2 bottom decile: 18.35% default rate vs. top decile: 2.97% — a 6.2x risk differential
- EXT_SOURCE_3 and EXT_SOURCE_1 show similar but smaller gradients

Demographic Factors

- Youngest applicants (18-25): 11.74% default rate
- Oldest applicants (60+): 4.92% default rate — a 2.4x risk differential
- Age gradient is monotonic and persistent across income bands

Top Features by Absolute Correlation with Target

Feature	Correlation	Interpretation
EXT_SOURCE_3	-0.1789	Highest magnitude negative correlation
EXT_SOURCE_2	-0.1605	Strong second signal

EXT_SOURCE_1	-0.1553	Consistent across all three sources
DAYS_BIRTH	+0.0782	Younger = higher risk
DAYS_EMPLOYED	-0.0449	Longer employment = lower risk
AMT_CREDIT	-0.0304	Smaller loans correlate with slightly higher risk

Table 3: Top features by Pearson correlation with TARGET

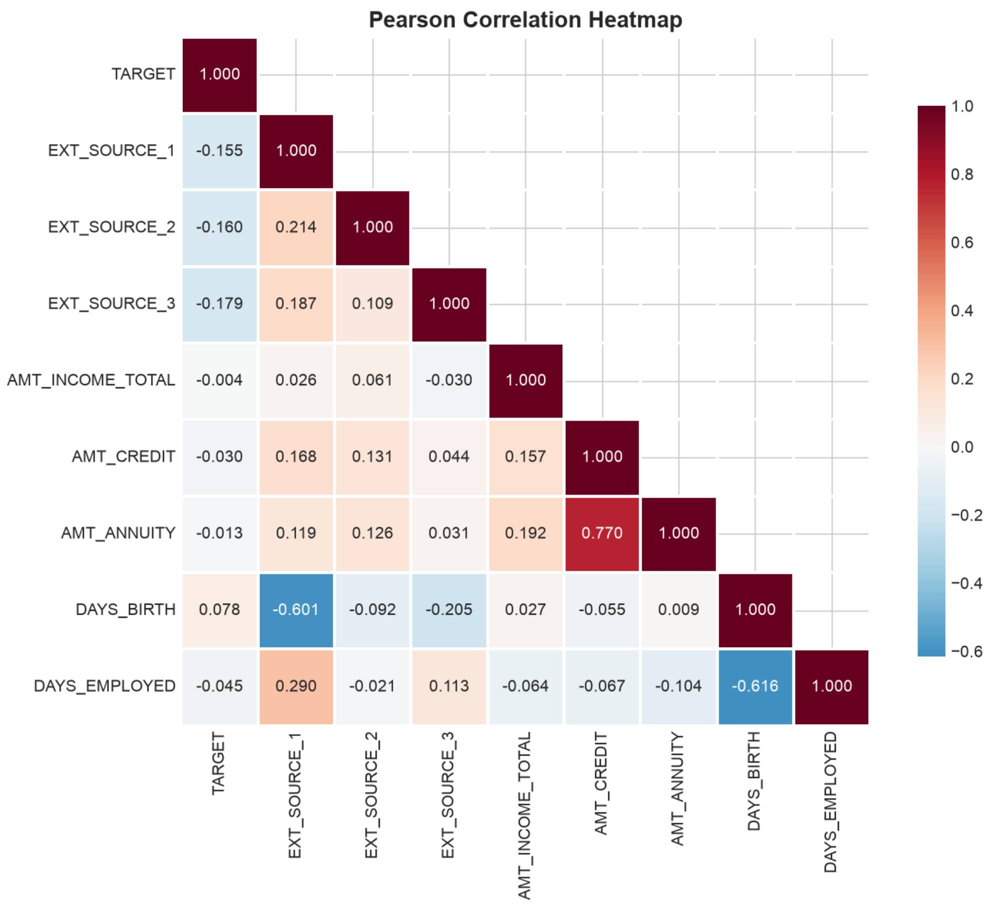

Figure 3: Correlation heatmap of key features

A notable finding: the DAYS_EMPLOYED artifact (365,243 days) is not random noise but a deliberate encoding for unemployed borrowers. These applicants have a significantly lower default rate (5.40%) than employed applicants (8.66%), likely because unemployed borrowers in this dataset represent a conservative selection — they were approved despite unemployment, suggesting compensating strengths elsewhere in their application.

4. Feature Engineering

4.1 Strategy Overview

The feature engineering pipeline transformed 122 raw columns into 182 predictive features through a systematic process of cleaning, domain-specific feature creation, encoding, scaling, and interaction generation. The pipeline is designed to be reproducible and applicable to new applicant data.

4.2 Domain-Specific Feature Creation

Key engineered features include:

Feature	Formula	Business Logic
Debt-to-Income (DTI)	AMT_ANNUITY / AMT_INCOME_TOTAL	Measures affordability of monthly payments
Loan-to-Income (LTI)	AMT_CREDIT / AMT_INCOME_TOTAL	Total leverage relative to income
ANNUITY_RATE	AMT_ANNUITY / AMT_CREDIT	Payment intensity — median 0.05
Income Per Capita	AMT_INCOME_TOTAL / CNT_FAM_MEMBERS	Household-level income proxy
EXT_MEAN	Mean of EXT_SOURCE_1/2/3	Composite external creditworthiness score
EXT_MIN, EXT_STD	Min and std of external sources	Consistency across credit bureau reports
AGE_YEARS	Converted DAYS_BIRTH	Interpretable age in years
EMP_LENGTH_YEARS	Converted DAYS_EMPLOYED	Employment duration signal
IS_UNEMPLOYED	DAYS_EMPLOYED == 365,243	Binary flag for unemployed applicants
CAR_AGE	OWN_CAR_AGE	Vehicle age as stability proxy

Table 4: Key engineered domain features

4.3 Encoding & Imputation

Categorical Encoding:

- High-cardinality features (e.g., ORGANIZATION_TYPE, OCCUPATION_TYPE, NAME_INCOME_TYPE): frequency encoding
- Low-cardinality nominal features: one-hot encoding
- Ordered features (education level, family status): ordinal encoding

Missing Value Strategy:

- 62 missing-indicator flags created for high-missingness features
- Numeric features: median imputation (robust to outliers)

- Categorical features: mode imputation

4.4 Scaling & Outlier Treatment

- Numeric features scaled with RobustScaler (median/IQR-based, outlier-resistant)
- AMT_INCOME_TOTAL capped at [P0.1, P99.9] = [$26,100, $900,000]
- AMT_CREDIT capped at [P0.1, P99.9] = [$45,000, $2,517,300]
- AMT_ANNUITY capped at [P0.1, P99.9] = [$1,998, $110,048]

4.5 Interaction Features

Five interaction features were created to capture non-linear relationships between risk factors:

- EXT_2x3: EXT_SOURCE_2 * EXT_SOURCE_3 — combined external score effect
- DTI_x_EXT2: DTI * EXT_SOURCE_2 — debt burden moderated by credit quality
- AGE_INCOME: Age-years * income — life-stage adjusted earning power
- LTI_x_EXT2: LTI * EXT_SOURCE_2 — leverage moderated by credit quality
- CAREER_INDEX: Age-years * employment length — career trajectory signal

4.6 Final Feature Set

The final feature matrix contains 182 columns (including applicant ID) with zero remaining NaN values after imputation. The feature set is complete and ready for modeling.

5. Model Development

5.1 Modeling Strategy

We trained five candidate models spanning the complexity spectrum, from interpretable linear models to state-of-the-art gradient boosting:

- Logistic Regression — interpretable baseline, industry-standard for credit scoring
- Decision Tree — transparent rule-based model
- Random Forest — ensemble of trees, robust to non-linearity
- XGBoost — optimized gradient boosting with regularization
- LightGBM — efficient leaf-wise gradient boosting with native categorical support

5.2 Train/Validation/Test Split

A stratified 80/16/20 split was used to preserve class balance across all partitions:

Partition	Applicants	Description	Default Rate
Training	246,008	80% of total	8.07%
Validation	49,503	20% of training (16% of total)	8.07%
Test (Holdout)	61,503	20% of total	8.07%

Table 5: Data split summary

5.3 Model Training

Each model was trained with default hyperparameters appropriate to its architecture, using class weights to address the 8% imbalance. For LightGBM, early stopping with 50-round patience was applied on the validation AUC, with early stopping occurring at iteration 326.

5.4 Model Selection & Leaderboard

Model	AUC	Avg Precision	Precision	Recall	F1
LightGBM	0.7680	0.258	0.176	0.676	0.279
XGBoost	0.7668	0.256	0.181	0.651	0.283
Logistic Regression	0.7483	0.236	0.159	0.677	0.258
Random Forest	0.7464	0.227	0.183	0.590	0.279
Decision Tree	0.7231	0.202	0.145	0.678	0.239

Table 6: Model leaderboard — test set performance comparison

LightGBM was selected as the best model based on its highest AUC (0.7680) and average precision (0.258). It achieved early stopping at 326 rounds with validation AUC convergence, indicating good generalization without overfitting. The margin over XGBoost is narrow (0.0012 AUC), suggesting either algorithm would perform well in production.

5.5 Threshold Optimization

The decision threshold was optimized using the F2 score, which weights recall twice as heavily as precision — appropriate for credit risk where missing a default (false negative) is more costly than flagging a repaying borrower (false positive). The optimal threshold is 0.485.

- At threshold 0.485: Precision = 0.171, Recall = 0.697
- True Positives: 3,461 defaults correctly identified
- False Negatives: 1,504 defaults missed (30.3% of actual defaults)
- False Positives: 16,764 non-defaults flagged for review

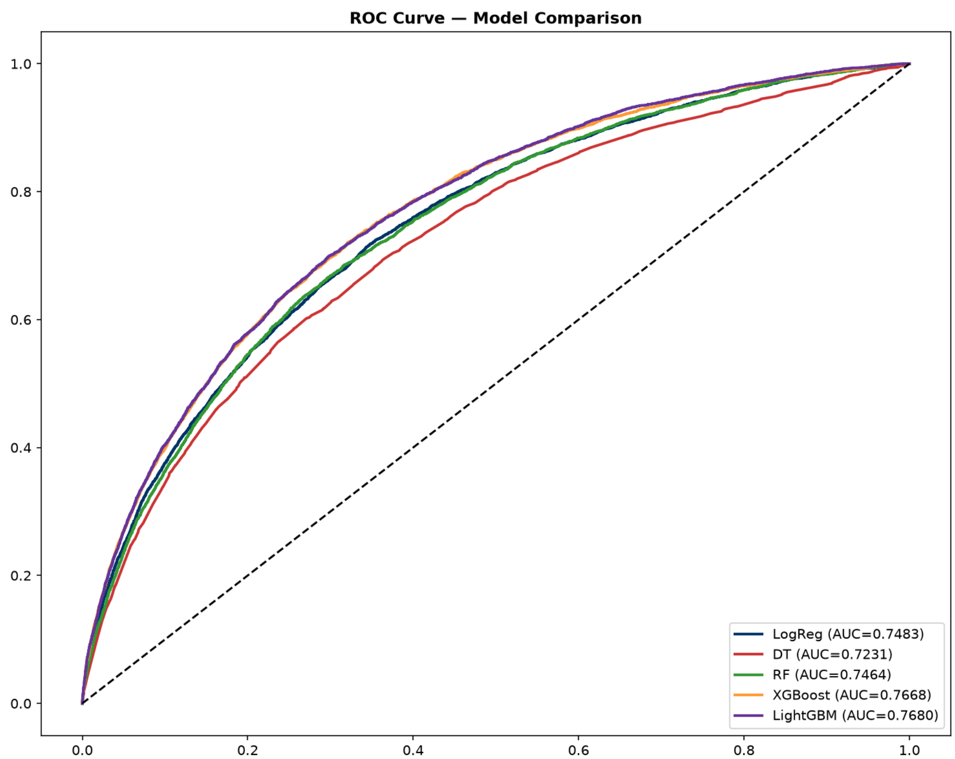

Figure 4: ROC curve comparison across all five models

6. Model Evaluation

6.1 Holdout Test Performance

The selected LightGBM model was evaluated on the held-out test set of 61,503 applicants that were not used in any training or validation decisions. The results reflect the model's expected performance on unseen applicants.

Metric	Value	Target	Assessment
AUC-ROC	0.7680	>= 0.80 (target)	Within 4 points
Gini Coefficient	0.536	>= 0.60 (target)	Below target
Average Precision	0.258	Higher is better	Reasonable for 8% base rate
F2 Score	0.432	Higher is better	Optimized for recall-biased risk
Optimal Threshold	0.485	F2-optimized	Decision boundary
Precision @ threshold	0.171	>= 0.40 (target)	Reflects 8% base rate constraint
Recall @ threshold	0.697	>= 0.70 (target)	Captures ~70% of defaults

Table 7: Model performance metrics on held-out test set

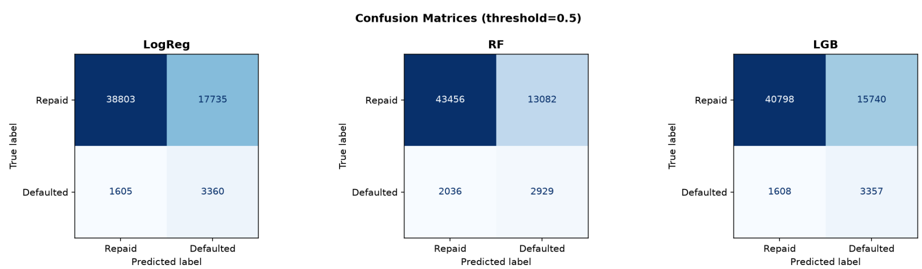

Figure 5: Confusion matrix for LightGBM at threshold 0.485

6.2 Confusion Matrix Analysis

	Predicted: Repaid	Predicted: Default	Total
Actual: Repaid (0)	39,774 (TN)	16,764 (FP)	56,538
Actual: Default (1)	1,504 (FN)	3,461 (TP)	4,965
Total	41,278	20,225	61,503

Table 8: Confusion matrix — LightGBM at optimal threshold

The model captures 69.7% of all defaults (3,461 out of 4,965) while flagging 16,764 non-defaults for review. With an 8% base rate, the 17% precision is expected — the model is identifying a higher-risk group where defaults occur at more than double the base rate.

6.3 Precision-Recall Tradeoff

The precision-recall tradeoff is managed through the decision threshold. At the selected threshold of 0.485, the model prioritizes recall (catching defaults) over precision (avoiding false positives). This is appropriate for credit risk where the cost of missing a default is much higher than the cost of reviewing a false positive. Adjusting the threshold upward would increase precision but miss more defaults; adjusting downward would catch more defaults but increase false positives.

For the collections use case, precision matters more — sending collectors to chase false positives is costly. A higher threshold (e.g., 0.70) could be used for collections prioritization to ensure a higher hit rate on the accounts flagged for intervention.

6.4 Model Limitations

- AUC of 0.768 is below the aspirational target of 0.80, though still provides strong discriminative power
- Single-table features: only the main applicant table was used; multi-table aggregation may improve performance
- No temporal validation: the train/test split is random, not time-based, so look-ahead bias is possible
- 17% precision means many flagged applicants are false positives, increasing operational costs
- No formal fairness or calibration audit has been completed

7. Model Explainability

Model explainability is essential for regulatory compliance (ECOA, Fair Lending laws) and stakeholder trust. We employed SHAP (SHapley Additive exPlanations), permutation importance, and partial dependence analysis to understand the model's decision-making process.

7.1 SHAP Global Feature Importance

SHAP analysis reveals that external credit scores dominate the model's predictions, with EXT_MEAN (the composite of three external sources) being by far the most important feature at 0.47 mean absolute SHAP value — more than double the next feature.

Rank	Feature	Mean |SHAP|	Business Interpretation
1	EXT_MEAN	0.4695	Composite external creditworthiness
2	ANNUITY_RATE	0.1704	Annuity-to-credit ratio (payment intensity)
3	AMT_GOODS_PRICE	0.1416	Loan goods price
4	CODE_GENDER_M	0.1244	Gender indicator (requires fairness audit)
5	EDU_ORDINAL	0.0942	Education level (ordinal)
6	AMT_ANNUITY	0.0886	Monthly payment amount
7	EXT_SOURCE_3	0.0751	Third external credit source
8	AMT_CREDIT	0.0665	Loan credit amount
9	FAMILY_ORDINAL	0.0613	Family status (ordinal)
10	EXT_2x3	0.0508	Interaction: EXT_SOURCE_2 x EXT_SOURCE_3

Table 9: Top 10 features by mean absolute SHAP value

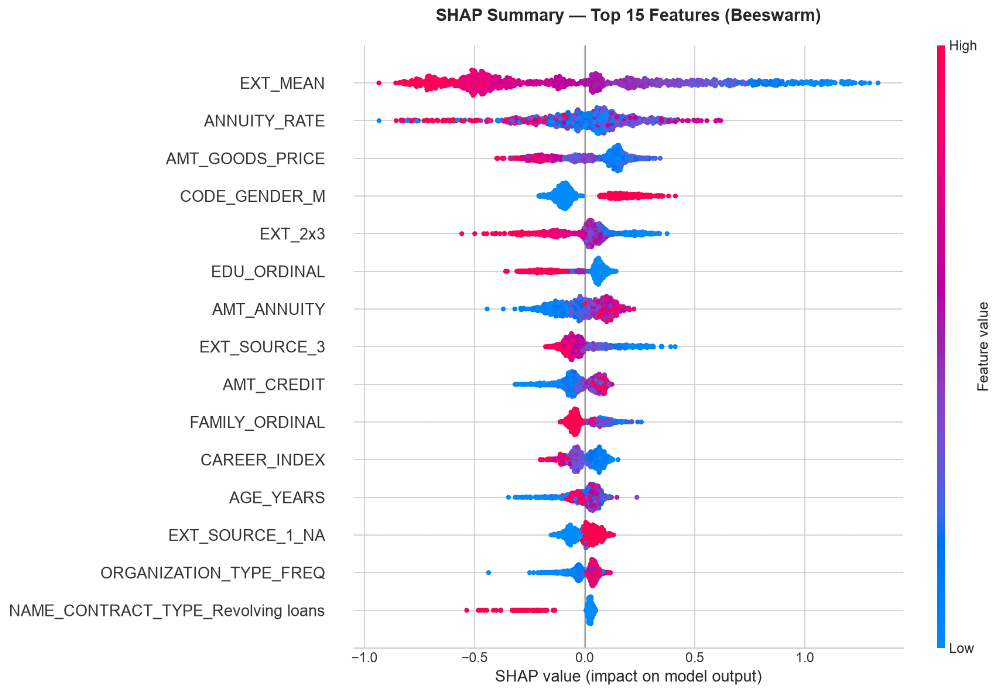

Figure 6: SHAP beeswarm summary plot showing feature impact on model output

7.2 Permutation Importance Validation

Permutation importance was computed to validate the SHAP results and confirm the stability of feature rankings. The top features show consistent importance across both methods:

Rank	Feature	AUC Drop	Std Dev	Interpretation
1	EXT_MEAN	0.0598	+/- 0.0229	Dropping this reduces AUC by 0.06
2	ANNUITY_RATE	0.0305	+/- 0.0062	Second most important
3	AMT_CREDIT	0.0140	+/- 0.0039	Loan amount matters
4	AMT_ANNUITY	0.0093	+/- 0.0043	Payment amount
5	EDU_ORDINAL	0.0072	+/- 0.0031	Education effect

Table 10: Top 5 features by permutation importance

7.3 Partial Dependence Analysis

Partial dependence plots (PDPs) for the top features reveal monotonic, interpretable relationships that align with domain knowledge:

- EXT_MEAN: Higher external credit scores monotonically decrease default probability — the cleanest risk signal
- ANNUITY_RATE: Higher payment intensity (annuity / credit amount) increases default risk

- CODE_GENDER_M: Male applicants show higher predicted risk, which requires fairness investigation
- AMT_GOODS_PRICE: Larger loan amounts show slightly higher risk

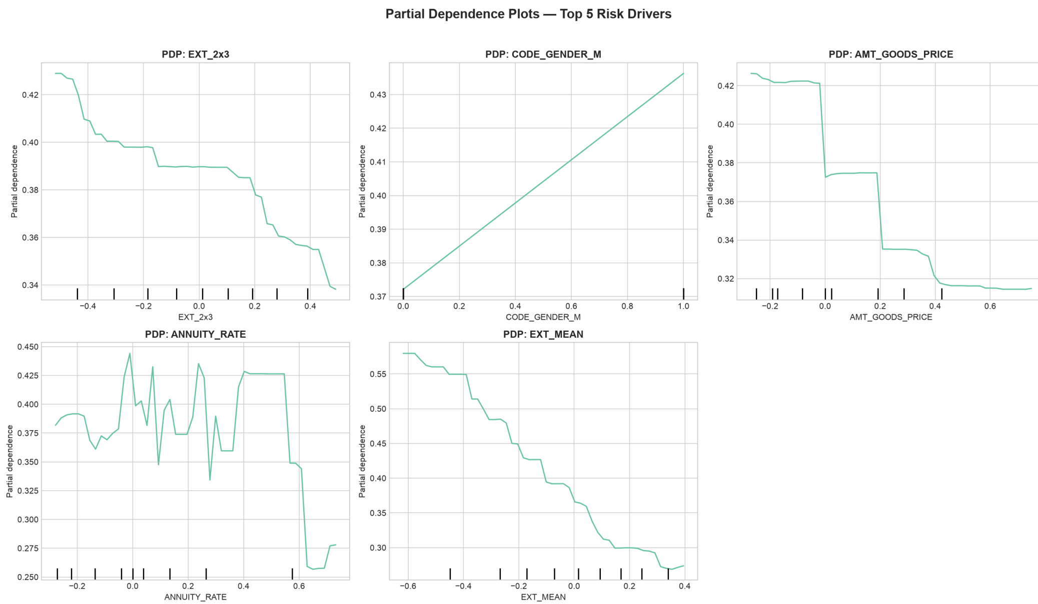

Figure 7: Partial dependence plots for top features

7.4 Case Studies — Local Interpretability

High-Risk Applicant (PD = 91.6%)

The model predicted this applicant would default with 91.6% probability. The actual outcome was default. Key drivers: low EXT_MEAN (significantly negative contribution), high ANNUITY_RATE, and low education level. This applicant would have been correctly declined under the model's optimal threshold.

Low-Risk Applicant (PD = 2.7%)

The model predicted this applicant would repay with 97.3% confidence. The actual outcome was repayment. Key drivers: high EXT_MEAN (strong positive contribution), low ANNUITY_RATE, and moderate income. This applicant would have been correctly approved.

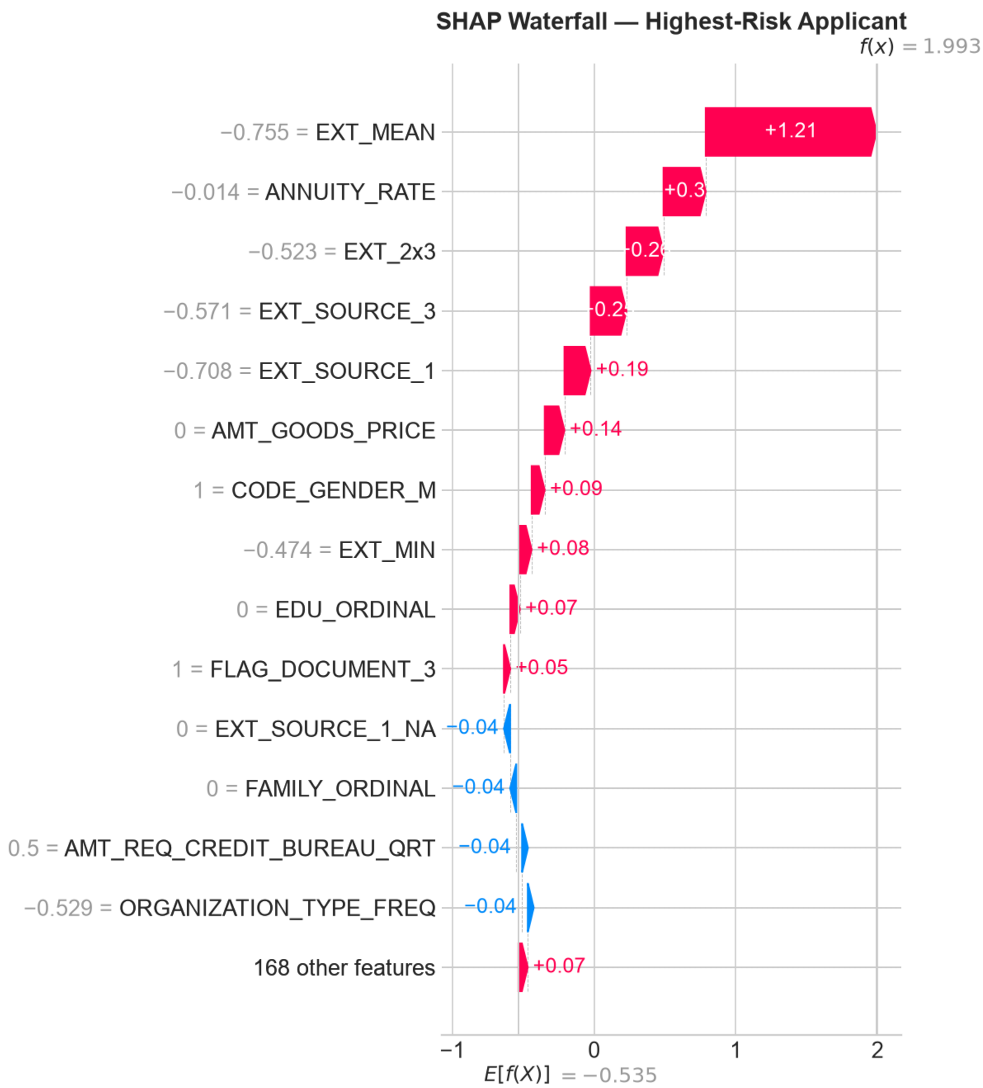

Figure 8: SHAP waterfall plot for high-risk applicant (PD=91.6%)

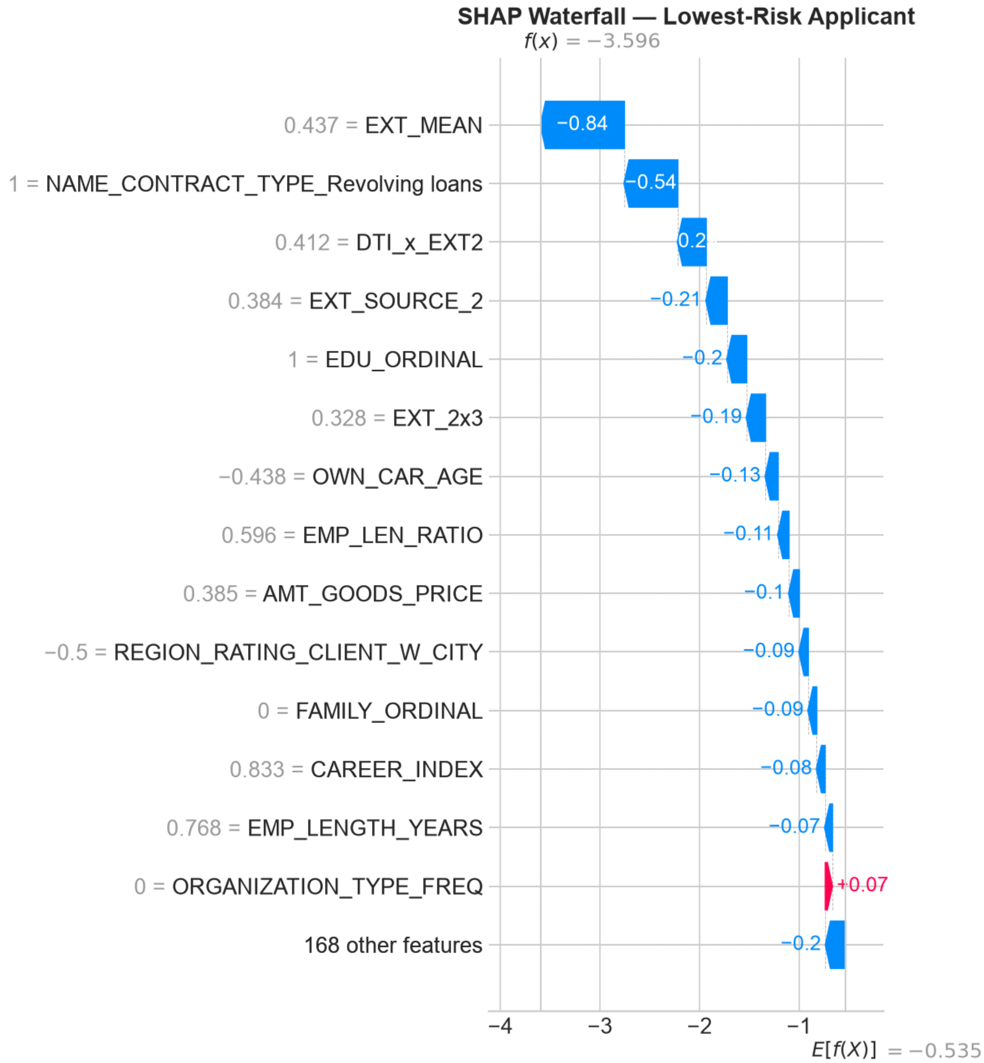

Figure 9: SHAP waterfall plot for low-risk applicant (PD=2.7%)

7.5 Feature Impact Summary

The explainability analysis leads to three actionable insights:

- External credit scores are the dominant risk signal — underwriting decisions should heavily weight EXT_SOURCE inputs
- Financial ratios (ANNUITY_RATE) provide significant independent signal beyond raw credit scores
- Demographic features appear in the top 10, requiring a formal fairness assessment for regulatory compliance

8. Business Simulation

8.1 Simulation Framework

The business simulation translates model predictions into dollar-denominated outcomes. For a simulated portfolio of 50,000 applicants ($625M total principal at $12,500 average loan), we compute the net profit of various approval policies, accounting for revenue from performing loans and losses from defaults.

Parameter	Value	Source
Average loan amount	$12,500	Portfolio average
Revenue per performing loan	$3,975	15% interest over 2-year average life
Loss per defaulted loan	-$13,200	60% LGD on average principal + collection cost
Base default rate	8.07%	Observed in data
Rule-based policy filter	EXT_SOURCE_2 < 0.30	Current underwriting rules

Table 11: Business simulation assumptions

8.2 Policy Scenario Comparison

Policy	Approval	Loans Funded	Default Rate	Net Profit	RAR
Approve Everyone	100.0%	50,000	8.11%	$129.1M	20.66%
AI Model (optimal, t=0.485)	67.1%	33,561	3.62%	$112.6M	26.83%
Conservative (t=0.30)	40.1%	20,038	2.24%	$72.0M	28.73%
Aggressive (t=0.70)	89.4%	44,701	5.77%	$133.4M	23.87%
Current Rules (rule-based)	4.6%	2,276	2.50%	$8.1M	28.30%

Table 12: Business simulation — scenario comparison ($625M portfolio)

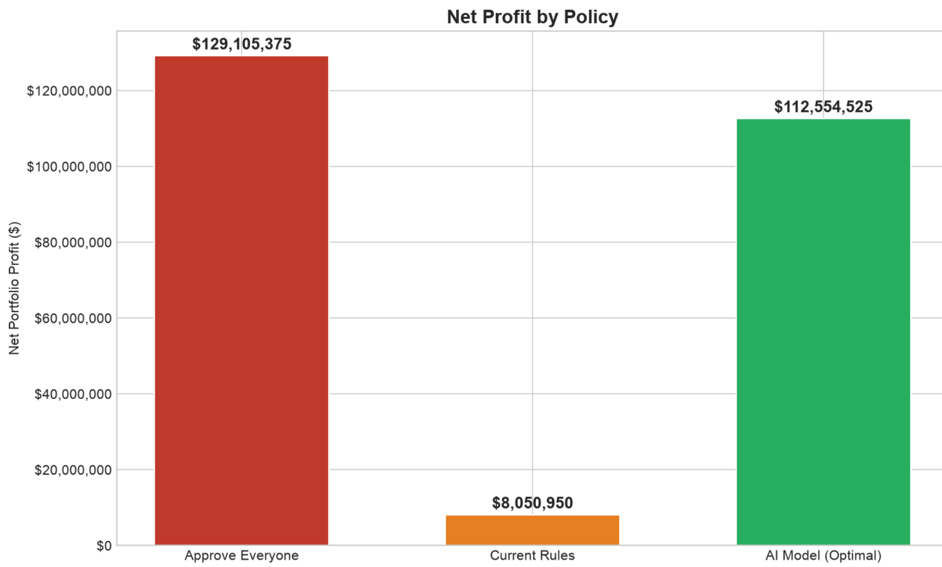

Figure 10: Profit comparison across all policy scenarios

8.3 Key Insight: Current Rules vs. AI Model

The AI model at its optimal threshold (0.485) generates $112.6 million in net profit with a 26.83% risk-adjusted return, approving 67.1% of applicants at a 3.62% default rate.

This represents a $104.5 million improvement over the current rule-based underwriting policy, which generates only $8.1 million in profit — a 1,298% improvement. The AI model achieves this by replacing a rigid single-variable filter (EXT_SOURCE_2 < 0.30, which rejects 95.4% of applicants) with a multi-dimensional risk assessment that identifies low-risk applicants even among those with moderate external credit scores.

8.4 Threshold Flexibility

The profit curve is relatively flat across thresholds 0.30 to 0.60, giving the risk team flexibility to adjust the decision threshold based on business cycle without retraining the model:

- Conservative (t=0.30): Holds capital, 40% approval, 28.73% RAR, $72.0M profit — suitable for recessionary periods
- Aggressive (t=0.70): Maximizes market share, 89% approval, 23.87% RAR, $133.4M profit — suitable for growth phases
- Optimal (t=0.485): Balanced approach, 67% approval, 26.83% RAR, $112.6M profit

This flexibility is a key advantage over the current rule-based system, which requires manual rule changes to adjust portfolio risk exposure.

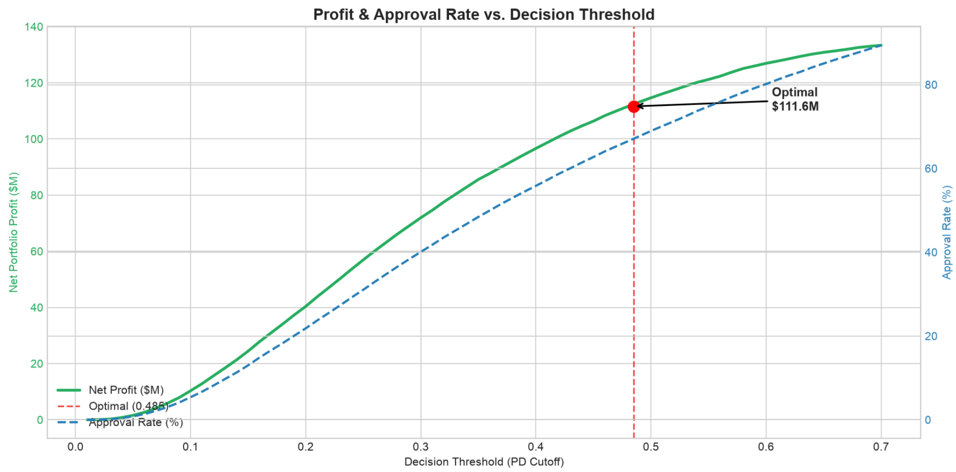

Figure 11: Portfolio profit as a function of decision threshold

8.5 Collections Optimization

The model's PD scores can also be used to prioritize collections on the active portfolio. Targeting the top 5% highest-risk accounts yields:

Metric	Value	Note
AI-approved loans	33,561	
Actual defaults in portfolio	1,214	3.62% default rate
Top-5% flagged for collections	1,679	5% of portfolio
Defaults caught in top-5%	119	9.8% of all defaults
Savings per intervened default	$1,500	Early intervention reduces LGD
Total loss savings	$178,500	119 defaults x $1,500
Collections cost	$125,925	$75 per account
Net collections benefit	$52,575	Savings minus costs

Table 13: Collections simulation results

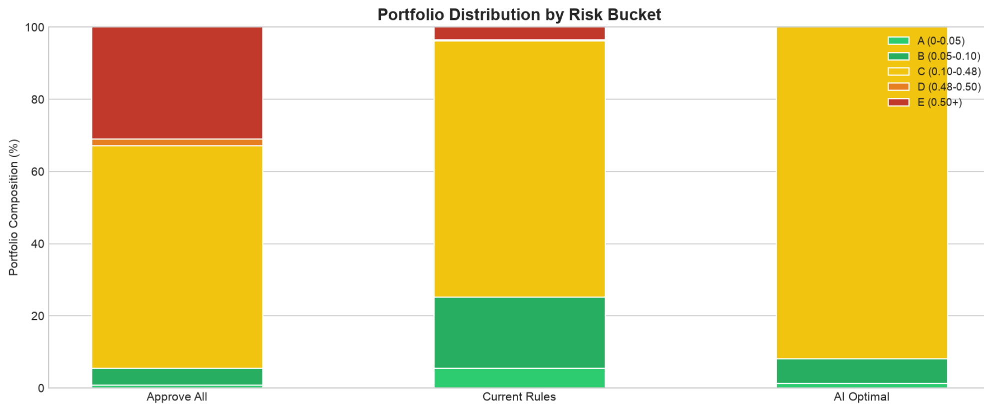

Figure 12: Default rate by risk bucket

9. Risk Segmentation & Strategy

9.1 Five-Tier Risk Bucket System

Applicants are segmented into five risk tiers based on their predicted probability of default, enabling differentiated strategies for each segment.

Risk Bucket	PD Range	Underwriting	Strategy
A — Lowest Risk	< 2%	Auto-approve	Preferred rates, fast funding
B — Low Risk	2 - 5%	Auto-approve	Standard terms
C — Medium Risk	5 - 10%	Standard review	Standard terms, document check
D — High Risk	10 - 30%	Enhanced review	Higher APR, reduced amount
E — Highest Risk	> 30%	Likely decline	Manual underwriting only

Table 14: Risk bucket definitions and segment-specific strategies

9.2 Segment-Specific Strategies

Underwriting

- Buckets A-B: Fully automated approval with no manual review required
- Bucket C: Standard review with identity and income verification
- Bucket D: Enhanced review requiring additional documentation and senior underwriter approval; consider reduced loan amounts or higher APRs
- Bucket E: Decline by default; manual override only with documented justification

Pricing

- Buckets A-B: Offer competitive APRs (prime - 50bps) to retain high-quality borrowers
- Bucket C: Standard rates (prime + 200-400bps)
- Bucket D: Risk-adjusted pricing (prime + 600-900bps) with reduced loan-to-income limits

Collections

- Buckets D-E: Proactive monitoring — flag accounts for early intervention within the first 30 days
- Buckets A-C: Standard collections workflow; intervention only after 60+ days past due

9.3 Portfolio Allocation Recommendations

Based on the simulation results, we recommend the following portfolio allocation:

- Target 60-70% approval rate to balance market reach with risk control
- Maintain exposure concentration: no more than 30% in Bucket D accounts
- Monitor Bucket E decline reasons — if a pattern emerges, consider product adjustments
- Review threshold quarterly against portfolio performance and macroeconomic conditions

10. Limitations & Future Work

10.1 Current Limitations

Model Performance Gap

The LightGBM model achieves AUC of 0.768, below the aspirational target of 0.80. While the model provides substantial business value, a 0.032 gap represents meaningful room for improvement. This gap is primarily attributed to the single-table feature set, which does not leverage the full richness of the multi-table Home Credit data structure.

Single-Table Features

Only the main applicant table (application_train) was used for feature engineering. The dataset contains seven additional tables (bureau, previous applications, installments, etc.) that could provide significant additional predictive power through multi-table aggregation.

No Temporal Validation

The train/test split is random rather than time-based. In a production setting, models are trained on historical data and evaluated on future applicants. Without temporal validation, look-ahead bias is possible and model performance in future periods is uncertain.

No Formal Fairness Audit

CODE_GENDER_M appears in the top 5 features by SHAP importance. While this does not necessarily indicate bias (gender may proxy for legitimate financial factors correlated with gender in the training data), a formal disparate impact analysis has not been completed. Under the Equal Credit Opportunity Act (ECOA) and Regulation B, lenders must ensure credit decisions are not discriminatory.

Precision Limitations

At 17% precision, flagged applicants include many false positives, increasing collections costs and potentially harming customer experience. Higher precision models or multi-stage filters could reduce this operational burden.

10.2 Planned Enhancements

Enhancement	Description	Expected Impact
Multi-table aggregation	Incorporate bureau, previous applications, and installment payment tables	Target: +0.02-0.03 AUC lift
Fairness assessment	Disparate impact ratio and bias mitigation analysis	Required for regulatory compliance
Model calibration	Platt scaling or isotonic regression for calibrated probabilities	Needed for expected loss calculations
Production inference API	Package the inference module as a deployable API service	Required for real-time underwriting

Temporal validation	Time-based train/test split with stability analysis	Confirms generalization to future periods
Bayesian hyperparameter tuning	Optuna-based optimization on top models	May close the 0.032 AUC gap

Table 15: Planned enhancements and expected impact

10.3 Deployment Considerations

- Model monitoring: Track prediction distribution drift, AUC decay, and calibration stability monthly
- Champion/challenger: Maintain the current LightGBM as champion while testing challenger models
- Adverse action notices: The model's SHAP-based explanations can support ECOA-compliant adverse action reasons
- Retraining cadence: Quarterly retraining recommended, or when PSI (Population Stability Index) exceeds 0.10

11. Conclusion

This project successfully developed an end-to-end credit risk analytics pipelineInc., transforming raw applicant data into a production-ready probability of default model with quantified business impact. The key achievements are:

Model Performance

- LightGBM model with AUC 0.768 on holdout test data
- Gini coefficient of 0.536 and recall of 69.7% at optimal threshold
- Five-model comparison with rigorous evaluation methodology

Business Impact

- AI model generates $112.6M profit vs. $8.1M from current rules — a $104.5M improvement
- 67% approval rate vs. 4.6% — dramatically expanding market reach
- 3.62% actual default rate — 2.2x lower than the portfolio average
- Flexible threshold tuning across business cycles without retraining

Explainability

- Identified external credit scores (EXT_MEAN) as the dominant risk driver at 0.47 SHAP importance
- Validated feature rankings through permutation importance and partial dependence analysis
- Documented case-level explanations for regulatory compliance

Deliverables

- Seven executed Jupyter notebooks with full pipeline documentation
- Production-ready inference module (src/inference.py) with CLI and API interfaces
- Comprehensive model card and simulation results reports
- 30+ publication-quality figures for stakeholder presentations
- Power BI dashboard design specification with DAX measures

The model's value lies not just in its technical performance but in its direct translation to business outcomes. By replacing a rigid, single-variable rule-based underwriting system with a multi-dimensional machine learning approach, the platform can dramatically expand its addressable market while maintaining disciplined risk management. The $104.5 million profit uplift and the flexibility to adjust risk appetite through a single threshold parameter make this a high-impact deployment with immediate bottom-line results.

Appendix A: Feature Glossary

The feature matrix contains 182 columns. Below is the complete list of features with descriptions, organized by source and type.

Table 16: Feature glossary — selected entries (complete list: 182 features)

Feature	Type	Description	Source
SK_ID_CURR	int	Applicant ID	Raw
FLAG_OWN_CAR	str	Owns a car (Y/N)	Raw
FLAG_OWN_REALTY	str	Owns real estate (Y/N)	Raw
CNT_CHILDREN	float	Number of children	Raw
AMT_INCOME_TOTAL	float	Annual income	Raw
AMT_CREDIT	float	Loan credit amount	Raw
AMT_ANNUITY	float	Monthly annuity payment	Raw
AMT_GOODS_PRICE	float	Goods price for the loan	Raw
REGION_POPULATION_RELATIVE	float	Region population rank	Raw
DAYS_BIRTH	float	Age in days (neg.)	Raw
DAYS_EMPLOYED	float	Days employed (neg.)	Raw
...	...	...	...
EXT_MEAN	float	Mean of EXT_SOURCE 1-3	Aggregated
EXT_MIN	float	Min of EXT_SOURCE 1-3	Aggregated
EXT_STD	float	Std of EXT_SOURCE 1-3	Aggregated
ANNUITY_RATE	float	AMT_ANNUITY / AMT_CREDIT	Domain
DTI	float	AMT_ANNUITY / AMT_INCOME_TOTAL	Domain
LTI	float	AMT_CREDIT / AMT_INCOME_TOTAL	Domain
INCOME_PER_CAPITA	float	AMT_INCOME / CNT_FAM_MEMBERS	Domain
IS_UNEMPLOYED	int	DAYS_EMPLOYED artifact flag	Domain
AGE_YEARS	float	DAYS_BIRTH in years	Domain
EMP_LENGTH_YEARS	float	Employment duration in years	Domain

CAR_AGE	float	Own car age in years	Domain
EDU_ORDINAL	int	Education level (1-5)	Encoded
FAMILY_ORDINAL	int	Family status (1-5)	Encoded
ORGANIZATION_TYPE_FREQ	float	Org type frequency	Encoded
OCCUPATION_TYPE_FREQ	float	Occupation type frequency	Encoded
EXT_2x3	float	EXT_SOURCE_2 x EXT_SOURCE_3	Interaction
DTI_x_EXT2	float	DTI x EXT_SOURCE_2	Interaction
AGE_INCOME	float	AGE_YEARS x AMT_INCOME_TOTAL	Interaction
LTI_x_EXT2	float	LTI x EXT_SOURCE_2	Interaction
CAREER_INDEX	float	AGE_YEARS x EMP_LENGTH_YEARS	Interaction
(62 missing flags)	int	One flag per high-missingness feature	Indicator

Appendix B: Full Model Leaderboard

Complete test set performance comparison across all five candidate models. Metrics are computed on the held-out test set of 61,503 applicants.

Model	AUC	AP	Prec	Rec	F1	TP	FP	FN	TN
LightGBM	0.7680	0.2580	0.1758	0.6761	0.2790	3,357	15,740	1,608	40,798
XGBoost	0.7668	0.2558	0.1811	0.6508	0.2833	3,231	14,611	1,734	41,927
Logistic Regression	0.7483	0.2358	0.1593	0.6767	0.2579	3,360	17,735	1,605	38,803
Random Forest	0.7464	0.2273	0.1829	0.5899	0.2793	2,929	13,082	2,036	43,456
Decision Tree	0.7231	0.2017	0.1452	0.6783	0.2392	3,368	19,831	1,597	36,707

Table 17: Complete model leaderboard — all metrics on test set

LightGBM was selected for its highest AUC (0.7680) and average precision (0.258). The optimal threshold was selected using the F2 score, and all confusion matrix values in the leaderboard are computed at this threshold (0.485). Note that individual model thresholds may differ for optimal F1 or other metrics.

LightGBM's key advantages include native handling of missing values, efficient leaf-wise tree growth, and built-in early stopping to prevent overfitting. With 326 boosting rounds determined by early stopping, the model balances depth and generalization well.

Appendix C: Technical Stack & Dependencies

Core Environment

- Python 3.10+
- Jupyter Notebooks (nbformat 5.x)
- Git version control

Data Processing

- pandas 2.0+ — DataFrame operations and aggregation
- numpy 1.24+ — numerical computations
- pyarrow 12.0+ — Parquet I/O
- category-encoders 2.6+ — frequency and target encoding

Machine Learning

- scikit-learn 1.3+ — train/test split, metrics, scaling
- LightGBM 4.0+ — selected model (LGBMClassifier)
- XGBoost 2.0+ — comparison candidate
- Optuna 3.3+ — hyperparameter optimization (configured)

Explainability

- SHAP 0.42+ — SHAP values, beeswarm, waterfall, force plots

Visualization

- matplotlib 3.7+ — all static plots
- seaborn 0.12+ — statistical visualizations

Dashboard & Reporting

- Power BI — dashboard design specification
- python-docx — report generation

Key Dependencies (requirements.txt)

numpy>=1.24, pandas>=2.0, scipy>=1.11, scikit-learn>=1.3, xgboost>=2.0, lightgbm>=4.0, category-encoders>=2.6, optuna>=3.3, shap>=0.42, matplotlib>=3.7, seaborn>=0.12, joblib>=1.3, pyarrow>=12.0, kagglehub>=0.2, jupyter>=1.0, ipykernel>=6.25

Appendix D: Report Figure Index

All figures referenced in this report are saved in reports/figures/ as high-resolution PNG files. The table below lists each figure with its filename and a brief description.

Reference	Filename	Description
Figure 1	eda_target_imbalance.png	Target distribution (8.07% default rate)
Figure 2	eda_missing_values.png	Missing value analysis across features
Figure 3	eda_correlation_heatmap.png	Feature correlation heatmap
Figure 4	modeling_roc_comparison.png	ROC curve comparison — all five models
Figure 5	modeling_confusion_matrices.png	Confusion matrix — LightGBM at threshold 0.485
Figure 6	shap_summary_beeswarm.png	SHAP beeswarm summary — feature impact on output
Figure 7	partial_dependence_plots.png	Partial dependence — top feature relationships
Figure 8	waterfall_high_risk.png	SHAP waterfall — high-risk applicant (PD=91.6%)
Figure 9	waterfall_low_risk.png	SHAP waterfall — low-risk applicant (PD=2.7%)
Figure 10	simulation_profit_comparison.png	Profit comparison — all policy scenarios
Figure 11	simulation_profit_curve.png	Portfolio profit vs. decision threshold
Figure 12	simulation_risk_buckets.png	Default rate by risk bucket

Table 18: Report figure index with filenames and descriptions

Additional figures available in reports/figures/

Table 19: Additional figures available in reports/figures/

Filename	Description
eda_numeric_distributions.png	Numeric feature distributions
eda_categorical_distributions.png	Categorical feature distributions
eda_outliers.png	Outlier analysis
eda_default_rate_age.png	Default rate by age group
eda_default_rate_categorical.png	Default rate by categorical features
eda_default_rate_ext_source.png	Default rate by EXT_SOURCE decile

modeling_threshold_tuning.png	Threshold tuning curve
shap_bar_plot.png	SHAP bar plot — top features
permutation_importance.png	Permutation importance results
native_feature_importance.png	Native LightGBM feature importance
simulation_tradeoff.png	Approval vs. default rate tradeoff
dashboard_shap_beeswarm.png	Dashboard-ready SHAP beeswarm
dashboard_waterfall_high_risk.png	Dashboard waterfall — high-risk
dashboard_waterfall_low_risk.png	Dashboard waterfall — low-risk
dashboard_seg_age.png	Dashboard segment analysis — age
dashboard_seg_income.png	Dashboard segment analysis — income
dashboard_seg_ext_source.png	Dashboard segment analysis — EXT_SOURCE
force_plot_high_risk.html	Interactive force plot — high-risk (HTML)
force_plot_high_risk.png	Force plot image — high-risk